

Calc AB: Integrating Powers of Sine and Cosine

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DIRECTIONS

Evaluate each integral or simplify the expression. For ODD powers: peel one factor and use Pythagorean substitution. For EVEN powers: apply the half-angle identities in the reference box below. Problems 12-15 review integration, limits, and derivatives.

ODD POWERS | Peel one factor, then substitute:

$$\sin^2 x = 1 - \cos^2 x$$

$$\cos^2 x = 1 - \sin^2 x$$

EVEN POWERS | Use half-angle identities:

$$\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$$

$$\cos^2 \theta = \frac{1}{2}(1 + \cos 2\theta)$$

1 Evaluate:

$$\int \cos^2(2x) dx$$

Answer: _____

2 Evaluate:

$$\int \sin^3(3x) dx$$

Answer: _____

3 Evaluate:

$$\int \cos^4 x dx$$

Answer: _____

4 Evaluate:

$$\int \sin^2\left(\frac{x}{2}\right) dx$$

Answer: _____

5 Evaluate:

$$\int \sec^2(4x) dx$$

Answer: _____

6 Evaluate:

$$\int \sin^2 x \cos x dx$$

Answer: _____

7 Evaluate:

$$\int 5\sin^2(2x) dx$$

Answer: _____

8 Evaluate:

$$\int \cos^3 x \sin x dx$$

Answer: _____

9 Evaluate:

$$\int_0^{\pi/2} \cos^3 x dx$$

Answer: _____

10 Evaluate:

$$\int_0^{\pi/2} \sin^2 x dx$$

Answer: _____

11 Evaluate:

$$\int \tan^2(4x) dx$$

Answer: _____

12 Given:

$f(x) = 3x^2 - 2$, $f(2) = 14$. Find $f(1)$.

Answer: _____

13 Evaluate:

$$\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h}$$

Answer: _____

14 Find $f'(x)$ if:

$$f(x) = \ln\left(\frac{e^{\sin 3x}}{\cos x}\right)$$

Answer: _____

15 Find dy/dx if:

$$y = \tan^{-1}(3x)$$

Answer: _____



Answer Key & Solutions

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TEACHER NOTES

For integrals 1-11, verify answers by differentiating. Problem 9 ans=2/3; Problem 10 ans=pi/4. Problem 12: integrate f'(x) to get f(x), use f(2)=14 to find C=2, then f(1)=9. Problems 13-15 test limit definition, log differentiation, and arctan derivative.

1 Evaluate:

$$\int \cos^2(2x) dx$$

$$= \frac{1}{2}x + \frac{1}{8}\sin(4x) + C$$

Even: $\cos^2(2x) = (1+\cos 4x)/2$; integrate term by term.

2 Evaluate:

$$\int \sin^3(3x) dx$$

$$= -\frac{1}{3}\cos(3x) + \frac{1}{9}\cos^3(3x) + C$$

Odd: peel $\sin(3x)$, write $\sin^2 = 1-\cos^2$; u-sub $u = \cos(3x)$.

3 Evaluate:

$$\int \cos^4 x dx$$

$$= \frac{3}{8}x + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C$$

Even: $\cos^4 x = [(1+\cos 2x)/2]^2$; expand, then apply half-angle again.

4 Evaluate:

$$\int \sin^2\left(\frac{x}{2}\right) dx$$

$$= \frac{1}{2}x - \frac{1}{2}\sin(x) + C$$

Even: $\sin^2(x/2) = (1-\cos x)/2$; integrate directly.

5 Evaluate:

$$\int \sec^2(4x) dx$$

$$= \frac{1}{4}\tan(4x) + C$$

Direct antiderivative of $\sec^2$; $u = 4x$ gives a factor of $1/4$.

6 Evaluate:

$$\int \sin^2 x \cos x dx$$

$$= \frac{1}{3}\sin^3 x + C$$

u-sub $u = \sin x$, $du = \cos x dx$. Gives $\int u^2 du = u^3/3$.

7 Evaluate:

$$\int 5\sin^2(2x) dx$$

$$= \frac{5}{2}x - \frac{5}{8}\sin(4x) + C$$

Even: $\sin^2(2x) = (1-\cos 4x)/2$. Multiply by 5 and integrate.

8 Evaluate:

$$\int \cos^3 x \sin x dx$$

$$= -\frac{1}{4}\cos^4 x + C$$

u-sub $u = \cos x$, $du = -\sin x dx$. Gives $-\int u^3 du = -u^4/4$.

9 Evaluate:

$$\int_0^{\pi/2} \cos^3 x dx$$

$$= \frac{2}{3}$$

Odd: peel $\cos x$, write $\cos^2 = 1-\sin^2$; $u = \sin x$; eval $[u - u^3/3]$ from 0 to 1.

10 Evaluate:

$$\int_0^{\pi/2} \sin^2 x dx$$

$$= \frac{\pi}{4}$$

Even: $\sin^2 x = (1-\cos 2x)/2$; eval $[x/2 - \sin(2x)/4]$ from 0 to $\pi/2$.

11 Evaluate:

$$\int \tan^2(4x) dx$$

$$= \frac{1}{4}\tan(4x) - x + C$$

Identity: $\tan^2(4x) = \sec^2(4x) - 1$; integrate each term separately.

12 Given:

$$f(x) = 3x^2 - 2, f(2) = 14. \text{ Find } f(1).$$

$$= f(1) = 9$$

Integrate: $f(x) = x^3 - 2x + C$. Use $f(2)=14$ to find $C=2$. Then $f(1) = 9$.

13 Evaluate:

$$\lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$= \cos(x)$$

This is the limit definition of the derivative: $d/dx[\sin x] = \cos x$.

14 Find f'(x) if:

$$f(x) = \ln\left(\frac{e^{\sin 3x}}{\cos x}\right)$$

$$= 3\cos(3x) + \tan(x)$$

Simplify first: $f(x) = \sin(3x) - \ln(\cos x)$. Then differentiate each term.

15 Find dy/dx if:

$$y = \tan^{-1}(3x)$$

$$= \frac{3}{1+9x^2}$$

$d/dx[\arctan(u)] = u'/(1+u^2)$. Here $u = 3x$, $u' = 3$.

